NAME: ______________________ **PERIOD:** ________ **DATE:** __________

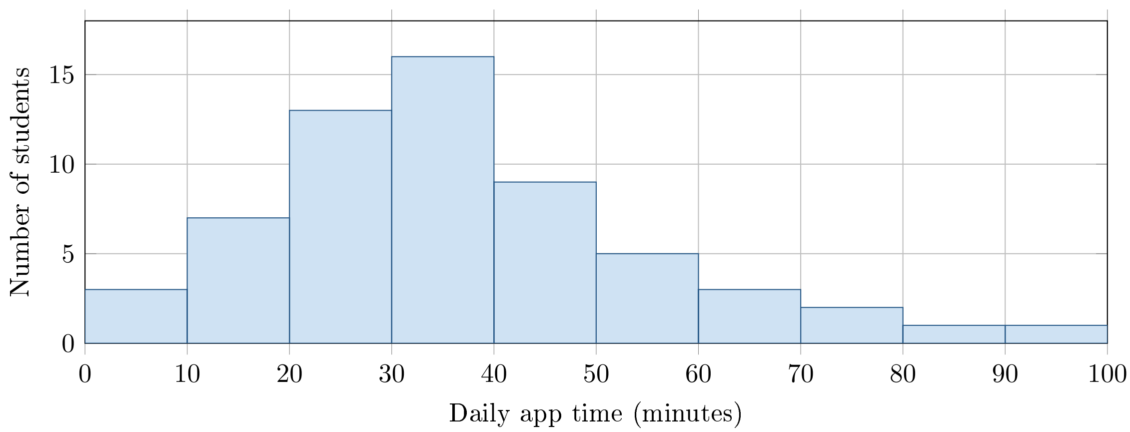
Check the Model

AP Statistics · Unit 2 · Topic 2.11 · B: 2026-11-12 · E: 2026-11-16

[STAR] TODAY'S PROBLEM

A school says daily app time is approximately normal with mean 40 minutes and standard deviation 12 minutes. The histogram shows 60 students; no recorded time was exactly 70 minutes. Table A gives a left area of 0.9938 for $z = 2.5$.

Ben: "About 16% should be above 52 minutes." **Ava:** "For 70 minutes, $z = 2.5$, so less than 1% should be above 70."



FIRST TAKE — before we discuss (5 min)

Do the times look balanced around 40 minutes, or does one side stretch farther? Point to the bars.

[BOOK] USING A NORMAL MODEL (keep on the page)

A normal curve is mound-shaped and symmetric, with population mean μ and standard deviation σ. Check whether the data's shape fits the model. About 68%, 95%, and 99.7% lie within 1, 2, and 3 standard deviations. Standardize with $z = (x - \mu)/\sigma$. Table A gives area to the left; right area is 1 minus left area. Observed percent is $100 \times \text{count}/\text{total}$. A model's calculated area need not match the observed data.

Work (a)–(c) from the rules box:

DOK 1 **(a)** State the proposed model's mean μ and standard deviation σ, with units.

DOK 2 **(b)** Check Ben's and Ava's model calculations. Then find the observed fraction of these 60 students above 70 minutes.

DOK 3 **(c)** Would you use Ben's and Ava's model results to describe these students? Use the histogram's shape and the observed versus predicted percent above 70. Explain what a reader could get wrong about long app times.

Frame: The calculations are __________ under the model; for these data I would __________ them because ____________________.

Frame: The model predicts ________ above 70, while the data show __________, so the model ______________ the upper tail.

EXIT — turn this in

One thing the rules box changed about my first take: ______________________________